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Generalized Jeffreys's approximate objective Bayes factor: Model-selection consistency, finite-sample accuracy, and the Jeffreys-Lindley Paradox in early-phase clinical trials

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Abstract.

Concerns about the misuse and misinterpretation of p-values and statistical significance have motivated alternatives for quantifying evidence. We define a generalized form of Jeffreys's approximate objective Bayes factor ($eJAB$), a one-line calculation that is a function of the p-value, sample size, and dimension of the test. We establish conditions under which $eJAB$ is model-selection consistent and verify them for commonly used statistical tests as well as through empirical analysis of several replication studies. We assess finite-sample approximation accuracy by comparing $eJAB$ with Markov chain Monte Carlo computed Bayes factors. Simulation studies compare the decision error probabilities of $eJAB$ with test-statistic Bayes factors (TSBF; Johnson, 2005; Johnson, Pramanik and Shudde, 2023), BIC approximations, and classical NHST. We also discuss a modification of $eJAB$ for one-sided tests and establish its model selection consistency. An exploratory reanalysis of 16,563 findings from 1,790 early-phase ClinicalTrials.gov trials yields 20 examples of the Jeffreys-Lindley paradox, where NHST (p-value ≤ 0.05) and Bayesian measures of evidence ($eJAB > 3$) are in strong disagreement. We report on these examples.

Key words and phrases: Jeffreys's approximate objective Bayes factor, Jeffreys–Lindley paradox, null-hypothesis significance testing, p-values, replicability.

1. INTRODUCTION

Concerns about the replicability of scientific findings, arising in part from the overreliance on null-hypothesis significance testing (NHST) at conventional p-value thresholds (Ioannidis, 2005), have prompted calls for more stringent standards of evidence (Benjamin et al., 2018a; Johnson, 2013), more judicious interpretation of statistical significance (Wasserstein and Lazar, 2016),

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and alternative measures for quantifying statistical evidence (Berger and Sellke, 1987; Evans, 2016; Grünwald, 2024; Halsey, 2019). One such measure is the Bayes factor—the ratio of marginal likelihoods under competing hypotheses (Kass and Raftery, 1995; Kass and Wasserman, 1995; Ly, Verhagen and Wagenmakers, 2016; Morey, Romeijn and Rouder, 2016; Raftery, 1995; Wagenmakers, 2007). The evidential behavior of p-values and Bayes factors has been studied through stochastic ordering in sample size and effect size, with implications for the Jeffreys–Lindley paradox, under which the two approaches can yield conflicting conclusions (Held and Ott, 2016, 2018; Huisman, 2023; Wagenmakers and Ly, 2023; Wei, Nathoo and Masson, 2023). An important practical consideration, however, is the computational complexity of Bayes factors. since they require integration over prior distributions, often via numerical methods such as Markov chain Monte Carlo (MCMC) with model-specific tuning (JASP, 2026; Morey and Rouder, 2026; Sinharay and Stern, 2005). Approximate Bayes factors offer a computationally convenient alternative when exact or full marginal-likelihood calculations are difficult. Such approximations can be derived from asymptotic theory, information criteria, p-values and test statistics, or summary statistics.

In an early but underappreciated contribution, Jeffreys (1936) developed an approximate Bayes factor, referred to here as JAB, for testing a point-null hypothesis against a composite alternative under a normal unit-information prior. The resulting expression contains the Wald statistic in an exponential term and the effective sample size in a square-root term. Through a detailed reexamination of Jeffreys's work, Wagenmakers (2022) further simplified the exponential component using a piecewise approximation to the quantile function of the χ^2_1 distribution. The finite-sample form depends on the prior centering; for the one-sided construction below, we use a symmetric local prior centered at the null value. A one-sided version of JAB can then be obtained by multiplying the two-sided JAB by a factor reflecting the posterior mass assigned to the theoretically predicted direction (Marsman and Wagenmakers, 2017). More recently, Wagenmakers, Bartoš and Aust (2026) showed that, for single-parameter Wald-type tests, certain forms of JAB can be interpreted as approximate e-processes. Because e-processes support sequential monitoring and optional stopping while controlling the probability of accumulating misleading evidence against the null hypothesis, this connection suggests that JAB may also possess useful approximate frequentist error-control properties in sequential settings.

Another influential approximation is based on the Bayesian information criterion (BIC) Kass and Wasserman (1995). For single-parameter tests, the Bayes factor approximation is asymptotically equivalent to JAB, with the likelihood-ratio test statistic replacing the Wald statistic. A related refinement is proposed by Bickel (2025), which incorporates a finite sample-size correction term into the BIC by raising the likelihood ratio to a correction-dependent power, thereby reducing evidential exaggeration under the null hypothesis.

An alternative line of research develops test-statistic Bayes factors (TSBFs) in closed form from the sampling distributions of test statistics under local alternatives, rather than treating them as approximations to full data Bayes factors (Johnson, 2005, 2008; Yuan and Johnson, 2008). More recently, the Bayes factor function has retained this core structure as a family of TSBFs evaluated over alternative hypotheses indexed by a standardized effect size, while extending the class of nonlocal priors used to define these alternatives (Datta et al., 2025a,b; Johnson, Pramanik and Shudde, 2023). For each fixed nonzero effect size, the resulting Bayes factor function is pointwise model-selection consistent.

Building on these developments, we generalize JAB, hereafter denoted $eJAB$, using the Savage–Dickey density ratio representation under an extension of the unit-information prior. This extension incorporates parameter dimension, accounts for finite-sample corrections, and facilitates applications to nonparametric testing. Our work differs from Bickel (2025) in its consideration of p-values beyond those from the likelihood ratio test and in our explicit statement of conditions for model-selection consistency. For the latter, Bickel (2025) demonstrates asymptotic equivalence to the BIC so that model-selection consistency arises under the conditions specified in Neath and Cavanaugh (2012). Our work also differs from Bartoš and Wagenmakers (2023), Rostgaard (2023), and Wagenmakers (2022) primarily in the formulation of the generalization and the development of its theoretical justification. We assess the finite-sample performance of $eJAB$ across ten common parametric and nonparametric tests. The results suggest that $eJAB$ is a broadly applicable and reliable approach to approximate the Bayes factor. We then apply $eJAB$ to 16,563 reported primary- and secondary-outcome findings from 1,790 early phase I, phase I, and phase II trials registered on https://ClinicalTrials.gov (CTG), accessed on August 7th, 2025. We report on 20 examples of the Jeffreys-Lindley paradox in clinical trials, where NHST (p-value ≤ 0.05) and the approximate Bayes factor ($eJAB > 3$) are in strong disagreement.

2. METHODOLOGY

DEFINITION 2.1. To test the hypotheses $\mathcal{H}_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ versus $\mathcal{H}_1 : \boldsymbol{\theta} \neq \boldsymbol{\theta}_0$, we define *eJAB* as

$$eJAB_{01} = \sqrt{n} \exp\left\{ -\frac{1}{2} \frac{n^{1/q} - 1}{n^{1/q}} Q_{\chi^2_q}(1-p) \right\},$$

where q is the dimension of the test, n is the sample size, $Q_{\chi^2_q}(\cdot)$ is the quantile function of the chi-squared distribution with q degrees of freedom (df), and p is the p-value from an NHST. We further define $eJAB_{10} = 1 \,/\, eJAB_{01}$.

DEFINITION 2.2. For a scalar right-sided test of $\mathcal{H}_0 : \theta = \theta_0$ versus $\mathcal{H}_+ : \theta > \theta_0$, let p_1 denote the one-sided p-value and define $p = 2\min\{p_1, 1-p_1\}$. The corresponding *eJAB* is given by

$$eJAB_{0+} = \frac{\sqrt{n}}{2(1-p_1)} \exp\left\{ -\frac{1}{2} Q_{\chi^2_1}(1-p) \right\}.$$

The left-sided test is handled analogously.

THEOREM 2.3. *Assuming the NHST p-value satisfies two regularity conditions,*

- ***R1****:* $p \xrightarrow{D} \text{Unif}(0,1)$ *as* $n \to \infty$ *under* $\mathcal{H}_0$,
- ***R2****:* $D_n = -\sqrt{n} \cdot p \ln p \xrightarrow{P} 0$ *as* $n \to \infty$ *under* $\mathcal{H}_1$,

then,

$$\operatorname*{plim}_{n\to\infty} eJAB_{01} = \begin{cases} \infty & \text{under } \mathcal{H}_0, \\ 0 & \text{under } \mathcal{H}_1. \end{cases}$$

The proof is provided in Supplementary Material, Appendix D.

When $q = 1$, the two *eJAB* definitions are related by

$$eJAB_{0+} = \frac{n^{\frac{1}{2(1-n)}}}{2(1-p_1)} eJAB_{01}^{\frac{n}{n-1}}.$$

Under $\mathcal{H}_0$, **R1** implies $eJAB_{01} \xrightarrow{P} \infty$. Since $n^{\frac{1}{2(1-n)}} \to 1$ and $(2(1-p_1))^{-1} \geq 1/2$, we have $eJAB_{0+} \xrightarrow{P} \infty$. Under $\mathcal{H}_+$, **R2** implies $eJAB_{01} \xrightarrow{P} 0$. Assuming additionally **R3**: $(1-p_1)^{-1} = \mathcal{O}_p(1)$ under $\mathcal{H}_+$, it follows that $eJAB_{0+} \xrightarrow{P} 0$. Therefore, the one-sided *eJAB* is also model-selection consistent.

FACT. We can derive *eJAB* as an approximation to the Bayes factor using a data-dependent prior that is a generalization of the unit-information prior for the parameters under test,

$$\boldsymbol{\theta} \sim \boldsymbol{\mathcal{N}}_q\left(\hat{\boldsymbol{\theta}},\ n^{1/q} \cdot \mathbf{J}_n^{-1}(\hat{\boldsymbol{\theta}})\right),$$

where $\hat{\boldsymbol{\theta}}$ is the MLE, and $\mathbf{J}_n(\hat{\boldsymbol{\theta}})$ is the observed information matrix of size $q \times q$. Under an asymptotic Gaussian approximation, the posterior density takes the form, $\pi(\boldsymbol{\theta} \mid \boldsymbol{y}, \mathcal{H}_1) \approx$

$$(2\pi)^{-\frac{q}{2}} |\mathbf{J}_n(\hat{\boldsymbol{\theta}})|^{\frac{1}{2}} \exp\left\{ -\frac{1}{2} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^\top \cdot \mathbf{J}_n(\hat{\boldsymbol{\theta}}) \cdot (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \right\}.$$

Assuming that the prior density for any nuisance parameters under $\mathcal{H}_0$ is everywhere equal to the conditional prior density for those parameters under $\mathcal{H}_1$, given the null value of the parameter under test, the Bayes factor (BF) can be expressed as the Savage–Dickey density ratio (Mulder, Wagenmakers and Marsman, 2022),

$$BF_{01} = \frac{\pi(\boldsymbol{\theta}_0 \mid \boldsymbol{y}, \mathcal{H}_1)}{\pi(\boldsymbol{\theta}_0 \mid \mathcal{H}_1)} \approx \sqrt{n} \exp\left\{ -\frac{1}{2} \frac{n^{1/q} - 1}{n^{1/q}} W \right\},$$

where $\boldsymbol{y}$ represents the data, and W is the Wald statistic. When the p-value is obtained from a Wald test, substituting $W = Q_{\chi^2_q}(1-p)$ yields $eJAB_{01}$ in Definition 2.1.

The prior underlying the approximate BF interpretation generalizes the unit-information prior which corresponds to $q = 1$. For cases where $q > 1$, the $n^{\frac{1}{q}}$ factor in the prior variance yields $\sqrt{n}$ growth in the evidence for $\mathcal{H}_0$, so that under eJAB the rate of evidence accumulation with sample size towards $\mathcal{H}_0$ is independent of the dimension q. When $q > 1$ (F- and chi-squared tests) and effect sizes are small, this dimensionless growth avoids the overly conservative behavior of the BIC approximation to the Bayes factor in favor of the null hypothesis (as detailed in the `R` Markdown files in the GitHub Simulations directory), while retaining model-selection consistency under R1 and R2.

FACT. When the p-value is obtained from a likelihood ratio test and $q = 1$, *eJAB* reduces to the evidential BIC method of Bickel (2025). This approximation is derived by applying a finite-sample correction to the BIC approximation to the BF, which amounts to raising the likelihood ratio to the power $(n-1)/n$. Model-selection consistency for the evidential BIC follows from its asymptotic equivalence to BIC together with standard consistency results for BIC (Bickel, 2025; Neath and Cavanaugh, 2012). By contrast, the conditions of Theorem 2.3 are broadly applicable: they cover nonparametric tests, provided **R1** and **R2** hold.

FACT. Applying the directional correction of Wagenmakers (2022, pp. 20–22) yields Definition 2.2. Unlike the MLE-centered prior used in the two-sided *eJAB* (Definition 2.1, $q = 1$), we use a symmetric local prior centered at θ_0, thereby eliminating the $(n-1)/n$

factor in the exponent of $eJAB_{0+}$. The directional correction uses a prior probability of 1/2 and an approximate posterior probability of $1-p_1$ in the direction of the alternative. Supplementary Material Appendix C.5 provides implementation details for the clinical-trial analysis.

3. SIMULATION STUDIES

We conduct 12 simulation studies across ten parametric and nonparametric tests to evaluate the regularity conditions and finite-sample accuracy. For each design, 5,000 datasets are generated under the null hypothesis and the alternative hypothesis with small, medium, and large effect sizes across various sample sizes. The designs include:

1. Two-sample t-test;
2. Simple linear regression;
3. Simple logistic regression;
4. One-way analysis of variance (ANOVA);
5. One-way repeated-measures ANOVA (rANOVA) with high and low intraclass correlations $\rho \in \{.9, .2\}$;
6. Chi-squared tests for independence under multinomial (total count fixed) and product-multinomial (row counts fixed) designs;
7. Cox proportional-hazards regression with right censoring;
8. Wilcoxon signed-rank test;
9. Mann–Whitney U-test;
10. Kruskal–Wallis H-test.

For nonparametric tests (items 8 through 10), we simulated using right-skewed and heavy-tailed (non-Gaussian) distributions. Detailed simulation settings, additional designs (e.g., two-way ANOVA and large contingency tables), and the `R` code for the subsequent subsections are available in the GitHub repository (see files within 'Theorem', 'Simulations' [parametric], and 'Supplementary Information' [nonparametric]).

We used p-values from the t-test for items 1 and 2; the F-test for items 4 and 5; the Wald test for items 3 and 7; and the asymptotic null distribution for item 6. The p-values for 8–10 are based on asymptotic distributions. The effective sample size n was taken as the number of uncensored observations for survival analysis (Volinsky and Raftery, 2000) and as the number of independent observations (i.e., the product of the number of participants and one less than the number of conditions) for rANOVA (Masson, 2011; Nathoo and Masson, 2016). For the other designs, n always refers to the total number of observations or counts. Among the tests with multiple df ($q > 1$) are items 4 through 6, and 10.

3.1 Verifying the Regularity Conditions

Condition **R1** of Theorem 2.3 is satisfied whenever the p-value is computed from an exact or asymptotic null distribution with a continuous cumulative distribution function.

It is not difficult to verify that **R2** is insufficiently strong to ensure model-selection consistency of the BIC approximation to the BF,

$$BF_{01}^{(\mathrm{BIC})} = n^{\frac{q}{2}} \exp\left\{-\frac{1}{2} Q_{\chi^2_q}(1-p)\right\}.$$

Using the asymptotic expansion of the incomplete gamma function (e.g., Daouia, Stupfler and Usseglio-Carleve, 2025, Lemma A.2), $\frac{1}{2}Q_{\chi^2_q}(1-p) = -\ln p + (\frac{q}{2}-1)\ln(-\ln p) - \ln\Gamma(\frac{q}{2}) + o(1)$. It follows that $BF_{01}^{(\mathrm{BIC})} \approx \Gamma(\frac{q}{2})\, n^{\frac{q}{2}} \cdot p\,(-\ln p)^{1-\frac{q}{2}}$, which requires a substantially stronger condition than **R2** for model-selection consistency when $q > 1$.

Next, we computed D_n of **R2** using the p-values and sample sizes from the simulated data, generated with medium effect sizes, across the ten designs, plus conditional logistic regression which is used in our application. In this subsection, items 4 and 5 ($\rho = .9$) each have $4-1=3$ df; item 6 has $(3-1)\times(3-1)=4$ df under a multinomial design; item 7 represents a semi-parametric Cox model; item 10 has $5-1=4$ df. The simulation results showing the distribution of D_n are displayed in Supplementary Figure S4 (Appendix E.1). Due to randomness, the x-axis of the figure has to display the total number of observations or pairs, rather than the number of uncensored observations for item 7, or the number of matched pairs for conditional logistic regression, as required in D_n and $eJAB$. Condition **R2** appears to hold for all fixed alternatives we examined. For local alternatives of the form $\theta^* = \theta_0 + c/n^\kappa$ in the one-sample t-test, **R2** does not hold at $\kappa = 1/2$ but does hold at $\kappa = 1/4$ (Supplementary Figure S4C). This follows from the denominator of the t-statistic being $\mathcal{O}_p(1/\sqrt{n})$.

In addition to simulations, we have also examined the feasibility of **R2** empirically, using realized values of D_n from four large-scale replication projects for results whose original study and *every* replication attempt reached $p \le 0.005$, for which we assume $\mathcal{H}_1$ is true. This allows us to observe realized values of D_n under $\mathcal{H}_1$ as a function of n. Appendix E.2 of the Supplementary Material describes the empirical study in detail; the realized values of D_n versus n are shown in Supplementary Figure S5. A clear pattern emerges, indicating that the distribution of D_n is shifted toward zero as n increases. The observed value of D_n at the largest value of n, $n = 49{,}196{,}731$ is $D_n^{(obs)} = 4.09 \times 10^{-29}$, and is circled in the figure. Overall, the study suggests that **R2** is practically feasible in real data.

3.2 Accuracy of the Bayes-Factor Approximation

To assess the finite-sample accuracy of an approximation to the BF, we compared *eJAB* with alternative computations in seven studies based on the parametric designs, using fixed representative sample sizes. In this subsection, item 7 represents a parametric survival regression model with an exponential baseline and expected censoring fractions of 23% under the null, and 21%, 19%, and 18% under increasing effect sizes. The MCMC-based BFs were computed by sampling the marginal posterior density using '*rstan*' (Stan, 2025), applying logspline density estimation (Kooperberg and Stone, 1992), and calculating the Savage–Dickey density ratio. In these cases, the model underlying the MCMC sampler assumed the same prior as *eJAB*, so discrepancies reflect errors of approximation rather than differences in prior.

Pairs of $(\ln eJAB_{10},\ \ln BF_{10})$ are color- and shape-coded by the p-value. Overall, *eJAB* tracks the MCMC-based BFs closely in Figure 1. Some inaccuracy, $\ln eJAB_{10} < \ln BF_{10}$, appears for logistic regression when evidence favors $\mathcal{H}_1$. Replacing *eJAB* with the BIC approximation to the BF performs well when $q = 1$, but marked severe discrepancies for $q > 1$, indicating that *eJAB* better approximates BF for ANOVA and rANOVA.

In sum, *eJAB* provides an accurate approximation to Bayes factors across a range of designs and effect sizes for parametric tests.

3.3 Bayesian Hypothesis Tests Using Nonparametric Statistics

Classical nonparametric tests do not specify a sampling distribution for the data, so marginal likelihoods are not defined. As such, the Bayes factor can be defined based on the marginal likelihood of the test statistic rather than on the data (Yuan and Johnson, 2008). We compared *eJAB* with Bayes factors constructed from nonparametric statistics (*BFNP*) across items 8 through 10. Figures depicting these results, a detailed description of the experiments, and the associated `R` Markdown files are available in the GitHub repository (see the 'Supplementary Information' directory).

Under $\mathcal{H}_1$, when effects and samples are both relatively small, *BFNP* tends to provide stronger evidence for $\mathcal{H}_1$ than *eJAB*. With larger effects or larger samples, the two methods produce comparable results. Under $\mathcal{H}_0$, *eJAB* effectively quantifies evidence for $\mathcal{H}_0$.

3.4 Equally Weighted Decision Error Profiles

For each effect size θ and sample size n, we compare a simple null (effect zero) with a simple alternative (effect θ). Under 0–1 loss, the equally weighted decision error (EWDE) for decision rule d is

$$\mathrm{EWDE}_d(\theta, n) = \frac{\alpha_d(\theta, n) + \beta_d(\theta, n)}{2},$$

where α_d is the type I error probability under the null and β_d is the type II error probability at the specified alternative. Define $\Delta E = \mathrm{EWDE}_{\mathrm{comparison}} - \mathrm{EWDE}_{\mathrm{eJAB}}$. The profiles display the normalized EWDE comparison,

$$\mathrm{EWDE}_{\mathrm{norm}}(\theta, n) = \frac{\Delta E(\theta, n)}{M},$$

where $M = \max_{(\theta,n)\in\mathcal{G}} |\Delta E(\theta, n)|$ and $\mathcal{G}$ is the grid over which the comparisons are computed. Positive values favor *eJAB*; negative values favor the comparison rule.

Figure 2 displays $\mathrm{EWDE}_{\mathrm{norm}}(\theta, n)$ comparisons of eJAB with: (A) the $p \le .005$ NHST decision rule, Benjamin et al. (2018b); (B) the BIC approximation to the Bayes factor; (C) TSBF where the true standardized effect size is assumed to be known and the Bayes factor function is evaluated at the true value; (D) TSBF where the standardized effect size is consistently estimated as $\hat{\omega} = \sqrt{X^2/n}$, with X^2 the chi-squared statistic and n the sample size, and this estimate is used in the Bayes factor function. The BIC comparison uses $BF_{01}^{\mathrm{BIC}} = n^{q/2}\exp(-\Lambda/2)$, where Λ is the likelihood-ratio statistic, for the six parametric-likelihood tests with available comparisons; Cox regression and the three rank tests are included for the comparisons with NHST but excluded for comparisons with BIC. The comparisons with TSBF focus strictly on chi-square tests for contingency tables. All Bayes factor approximations (*eJAB*, BIC, TSBF) use $BF_{01} < 1$ as the decision rule when evaluating error probabilities. NHST retains the rule $p \le \alpha$. The standalone $p \le .05$ comparison is provided in Supplementary Figure S3 (Appendix B).

Comparing *eJAB* to the $p <= 0.005$ decision rule, for all 10 settings considered, we observe a region in the EWDE corresponding to effect sizes that are not too small or too large, where *eJAB* exhibits better performance. The same can be said for the comparisons with the BIC approximation, where the EWDE exhibits a similar region of better performance for *eJAB* when $q > 1$. When $q = 1$ the performance of the BIC and *eJAB* Bayes factor decision rules is very similar. The comparisons with TSBF show that TSBF will have equal or better performance for chi-square tests when the Bayes factor function is evaluated at the true standardized effect size. In practice this parameter is unknown. When the standardized effect size is estimated, the situation changes, and *eJAB* becomes competitive, in particular for $q > 1$ and as q increases. Figure 2 displays dashed curves of the form $\theta = c/\sqrt{n}$ that hold $\sqrt{n}\theta$ constant. They preserve the alternative distribution when it depends on n and θ only through that quantity.

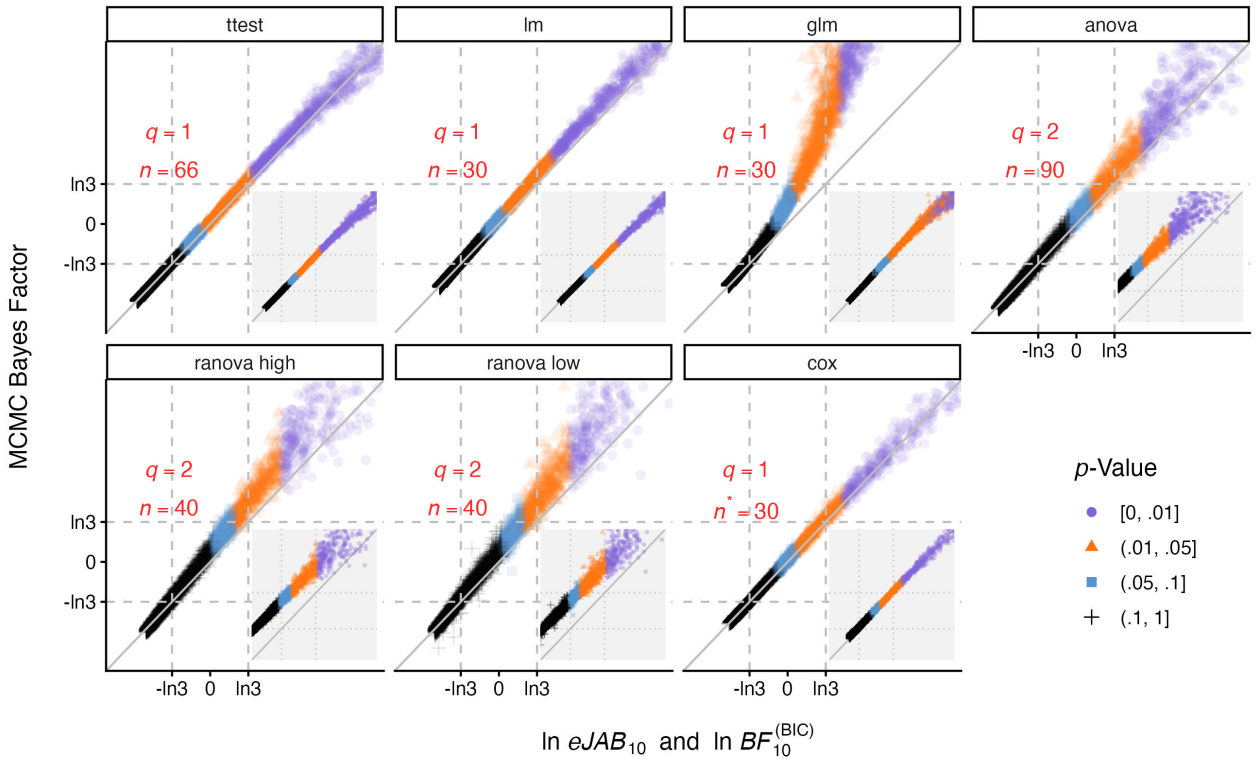


Fig 1. Comparison of the natural logarithms of $eJAB_{10}$ (main plots) and the BIC approximation (gray-background insets) with the MCMC-based Bayes factor. Red text indicates the test dimension q and effective sample size n, except for the Cox model, for which n^* denotes the number of observations rather than the number of events. Colors and shapes represent descriptive p-value intervals; NHST decisions use $p \leq \alpha$. Dashed lines indicate Bayes factors of 1/3 and 3.

4. THE JEFFREYS–LINDLEY PARADOX IN EARLY-PHASE CLINICAL TRIALS

We apply *eJAB* to ClinicalTrials.gov (CTG) results to illustrate the Jeffreys–Lindley paradox: a conventionally significant result can coexist with evidence favoring a point null under the specified Bayes-factor approximation.

We analyze 16,563 primary- and secondary-outcome CTG findings from 1,790 early-phase trials.

We computed the applicable *eJAB* from each finding's p-value, sample size, and test dimension (Definitions 2.1 and 2.2). NHST significance required $p_1 \leq .025$ for one-sided tests and $p \leq .05$ for two-sided/unrestricted tests. Preprocessing and exclusion criteria are described in Appendices A and C (Miron, Gonçalves and Musen, 2020).

Supplementary Figure S7B plots $\ln eJAB_{01}$ against the two-sided p-value, using $p = 2\min\{p_1, 1-p_1\}$ for one-sided tests. We screen for NHST significance in the specified test direction together with applicable eJAB > 3. For an appropriate point-null test with a numerically reported p-value, this is an illustration compatible with the Jeffreys–Lindley paradox under eJAB.

The screening catalogue includes 11 records with numerically reported p-values, all from five Phase II trials. Table 1 presents the ten largest calculated eJAB values among these records. The complete screening catalogue is provided in Supplementary Table S3 (Appendix C.7).

EJ: "The fact that such conflicts now go entirely unnoticed seems unproductive."

5. CONCLUSIONS

eJAB can be interpreted as an approximate objective Bayes factor when p-values are obtained from Wald or likelihood-ratio tests. The same interpretation should hold for tests whose statistics differ from the Wald statistic by a term that is $o_p(1)$. Simulations suggest that this interpretation holds more broadly. In addition, *eJAB* remains model-selection consistent for p-values from any test satisfying **R1** and **R2** of Theorem 2.3.

We compared the finite-sample performance of *eJAB* with Bayes factors computed from MCMC and the Savage–Dickey density ratio for the t-test, linear regression, logistic regression, ANOVA, rANOVA, and parametric survival analysis. Under the same generalization of the unit-information prior, *eJAB* closely approximates the MCMC-based Bayes factor. For nonparametric tests, namely Wilcoxon signed-rank, Mann–Whitney, and

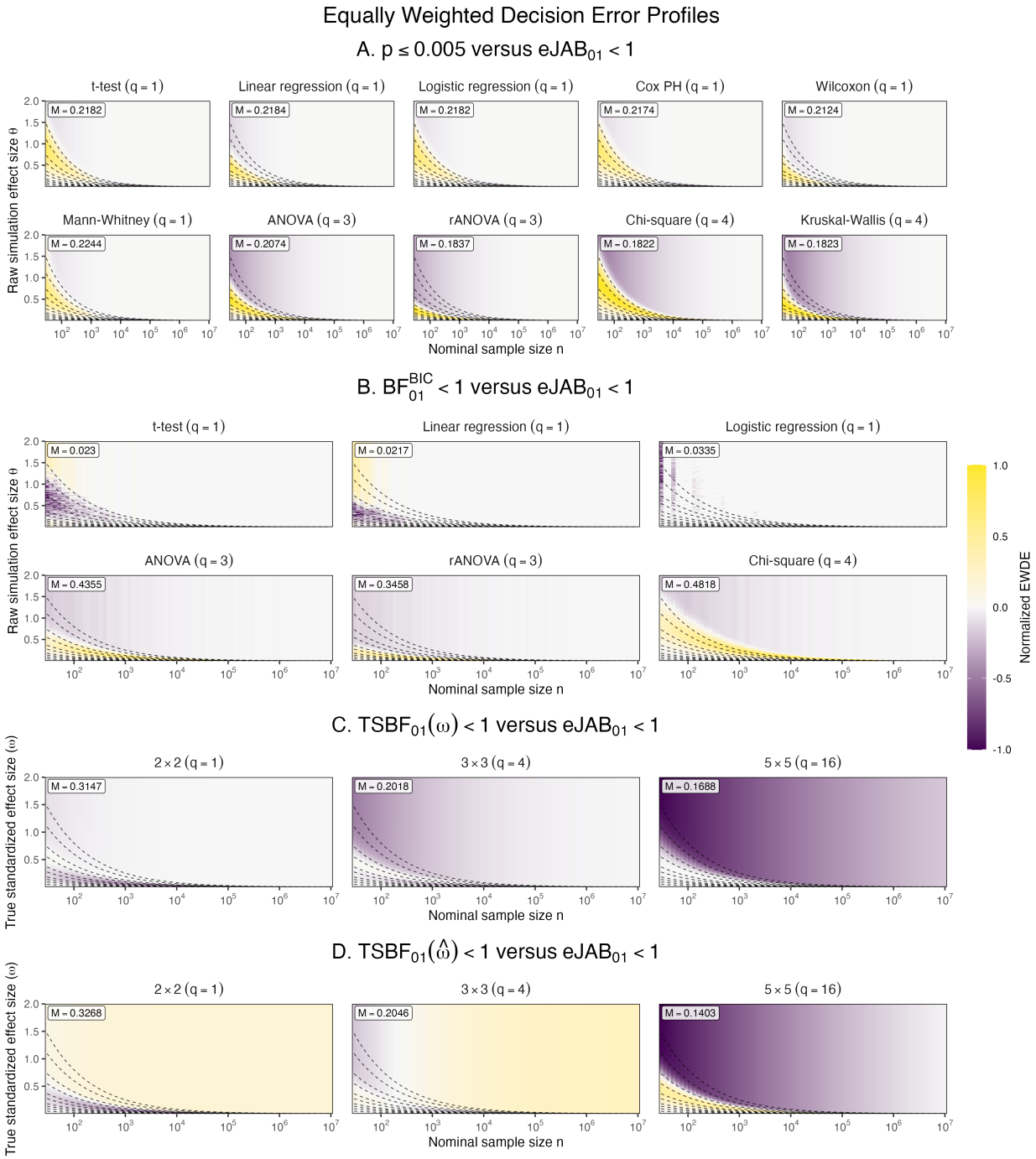


FIG 2. Equally Weighted Decision Error Profiles. (A) $p \leq .005$ versus $eJAB_{01} < 1$ across ten statistical tests. (B) $BF_{01}^{\mathrm{BIC}} < 1$ versus $eJAB_{01} < 1$ across six parametric-likelihood tests. (C) $TSBF_{01}(\omega) < 1$ versus $eJAB_{01} < 1$ and (D) $TSBF_{01}(\hat{\omega}) < 1$ versus $eJAB_{01} < 1$, with $\hat{\omega} = \sqrt{X^2/n}$, for 2×2, 3×3, and 5×5 contingency tables. Colors indicate normalized EWDE: positive values (yellow) favor eJAB, whereas negative values (purple) favor the comparison rule. Each panel reports its normalization constant M, the maximum absolute EWDE difference over its displayed grid. Dashed curves represent local alternatives of the form $\theta = c/\sqrt{n}$, with $\theta = \omega$ for the contingency-table comparisons.

TABLE 1
Ten largest calculated eJAB values among records in the Jeffreys–Lindley screening catalogue with numerically reported p-values, in descending order.

Outcome and analysis	p	eJAB
J05 NCT00565812		Phase II
Number of Participants With Joint Space Narrowing Progression *Logistic regression*; $n = 848$	0.047	**4.06**
J19 NCT05013008		Phase II
Plasma biomarker change: CCN family member 4 (O95388) *t-test*; $n = 921$	0.044	4.00
J20 NCT05013008		Phase II
Plasma biomarker change: TGF beta-1 proprotein (P01137) *t-test*; $n = 921$	0.044	4.00
J06 NCT01225822		Phase II
Major bleeding events: BIBR 1048 50 mg twice daily versus enoxaparin 40 mg daily *Logistic regression*; $n = 781$	0.0472	**3.91**
J18 NCT04376684		Phase II
Part 1: Percentage of Participants Alive and Free of Respiratory Failure at Day 28 *Logistic regression*; $n = 782$	0.0456	3.80
J09 NCT01225822		Phase II
VTE events and all-cause mortality: BIBR 1048 150 mg twice daily versus enoxaparin 40 mg daily *Logistic regression*; $n = 757$	0.0401	**3.36**
J10 NCT01225822		Phase II
VTE events: BIBR 1048 150 mg twice daily versus enoxaparin 40 mg daily *Logistic regression*; $n = 757$	0.0401	3.36
J08 NCT01225822		Phase II
Proximal DVT, pulmonary embolism and VTE-related mortality: BIBR 1048 225 mg versus 50 mg, both twice daily *Logistic regression*; $n = 753$	0.0373	3.15
J04 NCT00565812		Phase II
Number of Participants With Decrease in Total Analgesic Medication Use *Logistic regression*; $n = 971$	0.032	3.13
J03 NCT00405821		Phase II
HIV-1 Viral Load Difference Between Arms *t-test*; $n = 440$	0.05	3.09

Kruskal–Wallis, *eJAB* performs comparably to Bayes factors based on nonparametric test statistics. Our simulations also show improved performance compared to the BIC approximation to the Bayes factor when $q > 1$. Overall, the *eJAB* calculation provides a satisfactory Bayes factor approximation, it is broadly applicable, and its expression is very simple depending only on the p-value, sample size and the dimension of the test.

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SUPPLEMENTARY MATERIAL

Appendices A–G provide supplementary methods and results. The Online Supplement, data, and code are available in the GitHub repository. An online *eJAB* calculator is also available.

REFERENCES

BARTOŠ, F. and WAGENMAKERS, E.-J. (2023). A general approximation to nested Bayes factors with informed priors. *Stat* **12** e600. https://doi.org/10.1002/sta4.600

BENJAMIN, D. J., BERGER, J. O., JOHANNESSON, M., NOSEK, B. A., WAGENMAKERS, E.-J., BERK, R., BOLLEN, K. A., BREMBS, B., BROWN, L., CAMERER, C. et al. (2018a). Redefine statistical significance. *Nature Human Behaviour* **2** 6–10. https://doi.org/10.1038/s41562-017-0189-z

BENJAMIN, D. J., BERGER, J. O., JOHANNESSON, M., NOSEK, B. A., WAGENMAKERS, E.-J., BERK, R., BOLLEN, K. A., BREMBS, B., BROWN, L., CAMERER, C., CESARINI, D. et al. (2018b). Redefine statistical significance. *Nature Human Behaviour* **2** 6–10. https://doi.org/10.1038/s41562-017-0189-z

BERGER, J. O. and SELLKE, T. (1987). Testing a point null hypothesis: The irreconcilability of p values and evidence. *Journal of the American Statistical Association* **82** 112–122. https://doi.org/10.1080/01621459.1987.10478397

BICKEL, D. R. (2025). A small-sample Bayesian information criterion that does not overstate the evidence, with an application to calibrating p-values from likelihood-ratio tests. *Statistical Papers* **66** 1–17. https://doi.org/10.1007/s00362-025-01682-1

DAOUIA, A., STUPFLER, G. and USSEGLIO-CARLEVE, A. (2025). Corrected inference about the extreme Expected Shortfall in the general Max-Domain of Attraction. *Information and Inference: A Journal of the IMA* **14** iaaf023. https://doi.org/10.1093/imaiai/iaaf023

DATTA, S., GUHA, R., SHUDDE, R. and JOHNSON, V. E. (2025a). Bayes factors functions based on test statistics and non-local moment prior densities. *Statistics & Probability Letters* **219** 110330. https://doi.org/10.1016/j.spl.2024.110330

DATTA, S., GUHA, R., SHUDDE, R. and JOHNSON, V. E. (2025b). On Bayes factor functions. *Bayesian Analysis* **20** 1399–1427. https://doi.org/10.1214/25-BA1562

EVANS, M. (2016). Measuring statistical evidence using relative belief. *Computational and Structural Biotechnology Journal* **14** 91–96. https://doi.org/10.1016/j.csbj.2015.12.001

Grünwald, P. D. (2024). Beyond Neyman–Pearson: E-values enable hypothesis testing with a data-driven alpha. *Proceedings of the National Academy of Sciences* **121** e2302098121. https://doi.org/10.1073/pnas.2302098121

Halsey, L. G. (2019). The reign of the p-value is over: What alternative analyses could we employ to fill the power vacuum? *Biology Letters* **15** 1–8. https://doi.org/10.1098/rsbl.2019.0174

Held, L. and Ott, M. (2016). How the maximal evidence of p-values against point null hypotheses depends on sample size. *The American Statistician* **70** 335–341. https://doi.org/10.1080/00031305.2016.1209128

Held, L. and Ott, M. (2018). On p-values and Bayes factors. *Annual Review of Statistics and Its Application* **5** 393–419. https://doi.org/10.1146/annurev-statistics-031017-100307

Huisman, L. (2023). Are P-values and Bayes factors valid measures of evidential strength? *Psychonomic Bulletin & Review* **30** 932–941. https://doi.org/10.3758/s13423-022-02205-x

Ioannidis, J. P. A. (2005). Why most published research findings are false. *PLOS Medicine* **2** 696–701. https://doi.org/10.1371/journal.pmed.1004085

JASP (2026). JASP (Version 0.98.0)[Computer software].

Jeffreys, H. (1936). Further significance tests. *Mathematical Proceedings of the Cambridge Philosophical Society* **32** 416–445. https://doi.org/10.1017/S0305004100019125

Johnson, V. E. (2005). Bayes factors based on test statistics. *Journal of the Royal Statistical Society Series B: Statistical Methodology* **67** 689–701. https://doi.org/10.1111/j.1467-9868.2005.00521.x

Johnson, V. E. (2008). Properties of Bayes factors based on test statistics. *Scandinavian Journal of Statistics* **35** 354–368. https://doi.org/10.1111/j.1467-9469.2007.00576.x

Johnson, V. E. (2013). Revised standards for statistical evidence. *Proceedings of the National Academy of Sciences* **110** 19313–19317. https://doi.org/10.1073/pnas.1313476110

Johnson, V. E., Pramanik, S. and Shudde, R. (2023). Bayes factor functions for reporting outcomes of hypothesis tests. *Proceedings of the National Academy of Sciences* **120** e2217331120. https://doi.org/10.1073/pnas.2217331120

Kass, R. E. and Raftery, A. E. (1995). Bayes factors. *Journal of the American Statistical Association* **90** 773–795. https://doi.org/10.1080/01621459.1995.10476572

Kass, R. E. and Wasserman, L. (1995). A reference Bayesian test for nested hypotheses and its relationship to the Schwarz criterion. *Journal of the American Statistical Association* **90** 928–934. https://doi.org/10.2307/2291327

Kooperberg, C. and Stone, C. J. (1992). Logspline density estimation for censored data. *Journal of Computational and Graphical Statistics* **1** 301–328. https://doi.org/10.1080/10618600.1992.10474588

Ly, A., Verhagen, J. and Wagenmakers, E.-J. (2016). An evaluation of alternative methods for testing hypotheses, from the perspective of Harold Jeffreys. *Journal of Mathematical Psychology* **72** 43–55. https://doi.org/10.1016/j.jmp.2016.01.003

Marsman, M. and Wagenmakers, E.-J. (2017). Three insights from a Bayesian interpretation of the one-sided P value. *Educational and Psychological Measurement* **77** 529–539. https://doi.org/10.1177/0013164416669201

Masson, M. E. J. (2011). A tutorial on a practical Bayesian alternative to null-hypothesis significance testing. *Behavior Research Methods* **43** 679–690. https://doi.org/10.3758/s13428-010-0049-5

Miron, L., Gonçalves, R. S. and Musen, M. A. (2020). Obstacles to the reuse of study metadata in ClinicalTrials.gov. *Scientific Data* **7** 1–14. https://doi.org/10.1038/s41597-020-00780-z

Morey, R. D., Romeijn, J.-W. and Rouder, J. N. (2016). The philosophy of Bayes factors and the quantification of statistical evidence. *Journal of Mathematical Psychology* **72** 6–18. https://doi.org/10.1016/j.jmp.2015.11.001

Morey, R. D. and Rouder, J. N. (2026). BayesFactor: Computation of Bayes factors for common designs. R package version 0.9.12-4.8.

Mulder, J., Wagenmakers, E.-J. and Marsman, M. (2022). A generalization of the Savage–Dickey density ratio for testing equality and order constrained hypotheses. *The American Statistician* **76** 102–109. https://doi.org/10.1080/00031305.2020.1799861

Nathoo, F. S. and Masson, M. E. J. (2016). Bayesian alternatives to null-hypothesis significance testing for repeated-measures designs. *Journal of Mathematical Psychology* **72** 144–157. https://doi.org/10.1016/j.jmp.2015.03.003

Neath, A. A. and Cavanaugh, J. E. (2012). The Bayesian information criterion: Background, derivation, and applications. *WIREs Computational Statistics* **4** 199–203. https://doi.org/10.1002/wics.199

Raftery, A. E. (1995). Bayesian model selection in social research. *Sociological Methodology* **25** 111–163. https://doi.org/10.2307/271063

Rostgaard, K. (2023). Simple nested Bayesian hypothesis testing for meta-analysis, Cox, Poisson and logistic regression models. *Scientific Reports* **13** 1–11. https://doi.org/10.1038/s41598-023-31838-8

Sinharay, S. and Stern, H. S. (2005). An empirical comparison of methods for computing Bayes factors in generalized linear mixed models. *Journal of Computational and Graphical Statistics* **14** 415–435. https://doi.org/10.1198/106186005X47471

Stan (2025). RStan: the R interface to Stan. R package version 2.32.7.

Volinsky, C. T. and Raftery, A. E. (2000). Bayesian information criterion for censored survival models. *Biometrics* **56** 256–262. https://doi.org/10.1111/j.0006-341x.2000.00256.x

Wagenmakers, E.-J. (2007). A practical solution to the pervasive problems of p values. *Psychonomic Bulletin & Review* **14** 779–804. https://doi.org/10.3758/bf03194105

Wagenmakers, E.-J. (2022). Approximate objective Bayes factors from p-values and sample size: The $3p\sqrt{n}$ rule. *PsyArXiv* **1** 1–50. https://doi.org/10.31234/osf.io/egydq

Wagenmakers, E.-J., Bartoš, F. and Aust, F. (2026). Approximate Bayes factors are approximate e-processes. *SSRN*. https://doi.org/10.2139/ssrn.6138254

Wagenmakers, E.-J. and Ly, A. (2023). History and nature of the Jeffreys–Lindley paradox. *Archive for History of Exact Sciences* **77** 25–72. https://doi.org/10.1007/s00407-022-00298-3

Wasserstein, R. L. and Lazar, N. A. (2016). The ASA statement on p-values: Context, process, and purpose. *The American Statistician* **70** 129–133. https://dx.doi.org/10.1080/00031305.2016.1154108

Wei, Z., Nathoo, F. S. and Masson, M. E. J. (2023). Investigating the relationship between the Bayes factor and the separation of credible intervals. *Psychonomic Bulletin & Review* **30** 1759–1781. https://doi.org/10.3758/s13423-023-02295-1

Yuan, Y. and Johnson, V. E. (2008). Bayesian hypothesis tests using nonparametric statistics. *Statistica Sinica* **18** 1185–1200.